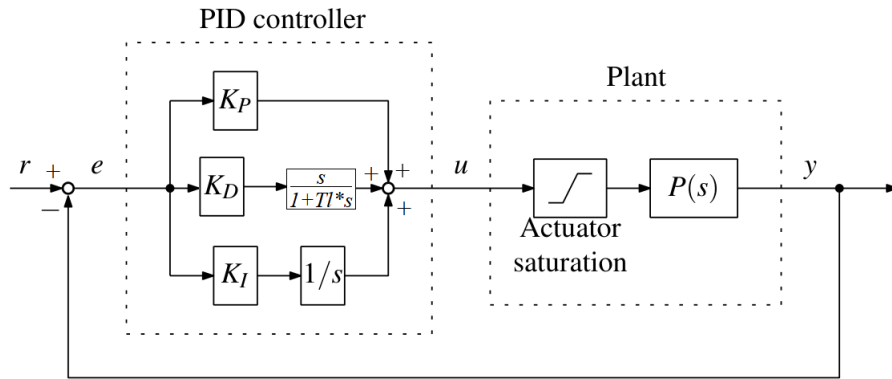


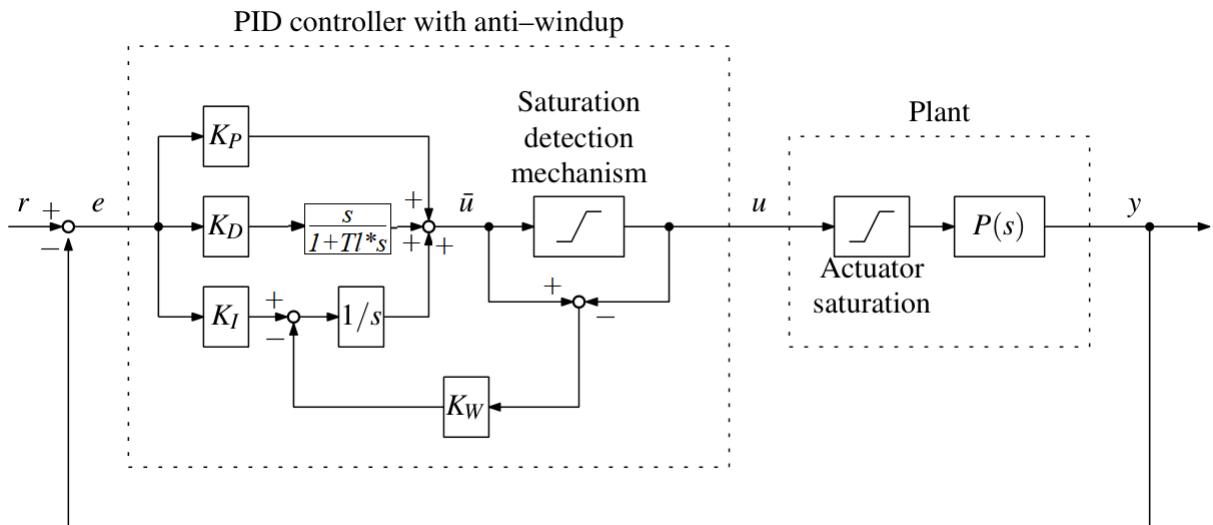
Controllore

1 PID



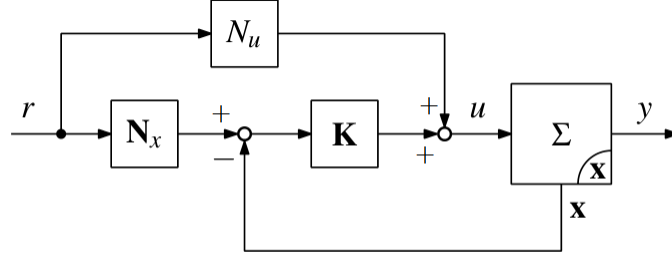
$$C(s) = K_P + \frac{K_I}{s} + K_D \frac{s}{1+T_I*s} = K_P \left(1 + \frac{1}{T_I s} + T_D \frac{s}{1+T_I*s} \right)$$

2 PID+Antiwindup



3 STATE-SPACE APPROACH

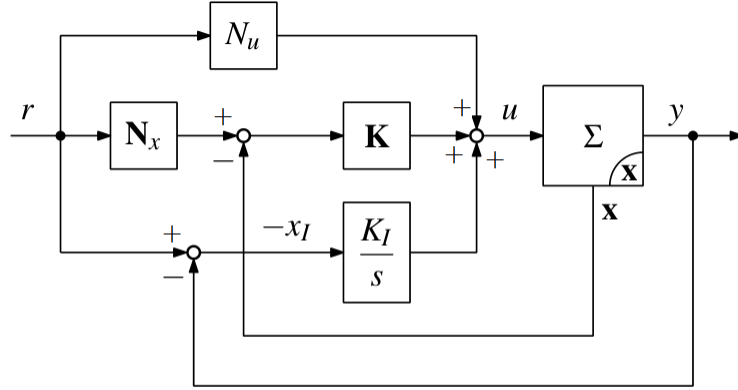
3.1 Nominal tracking



$$u = -\mathbf{K}\mathbf{x} + (N_u + \mathbf{K}N_x)r_\infty$$

3.2 Robust tracking

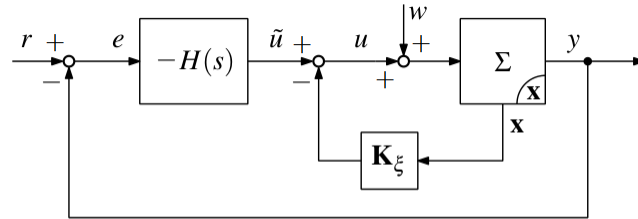
3.2.1 Robust tracking with integral action



$$u = u_\infty - \mathbf{K}(\mathbf{x} - \mathbf{x}_\infty) - K_I \int_0^t [y(\tau) - r(\tau)] d\tau$$

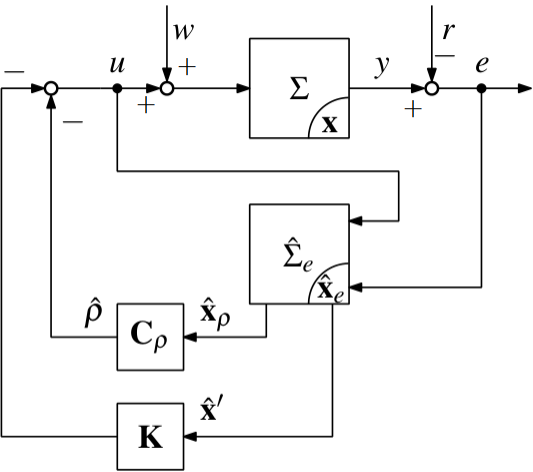
3.2.2 Robust tracking with internal model principle

3.2.2.1 Robust tracking with error-space approach



$$u = -\mathbf{K}_\xi \mathbf{x} - H(s)e$$

3.2.2.2 Robust tracking with extended estimator approach



$$u = -C_\rho \hat{x}_\rho - K \hat{x}'$$